DECISION ENGINEERING

Decision Card

One card per decision. Write the decision as a question with allowed answers, then record how each part of it is made, who holds the authority and what evidence it needs. Name the agents last.

DECISION ID	ACCOUNTABLE OWNER	RIGOR LEVEL	VERSION	REVIEW DATE
		☐ Basic ☐ Enhanced ☐ High		

1 Decision problem

The question and its allowed answers. Trigger and how often it occurs. Inputs and how current they must be. What is uncertain. The objective, the limits and the deadline.

2 Kinds

Which kind each part of the decision is, what makes that part, what it produces, and the rule, model or procedure it relies on. Several kinds can apply.

3 Authority

Who or what recommends, authorizes and executes. The permitted actions. Who can veto or escalate. Record influence separately: advice can still drive the outcome.

4 Coordination

Who must contribute. Which other decisions this one depends on or shares a limit with. How agreement is reached, and what happens when it is not.

5 Assurance

Evidence that the decision followed its rules (permission). Separate evidence that it achieved the intended result against a baseline over an agreed period (performance). The record kept, the fallback, and how changes to rules, models and prompts are approved.

6 Operating regime

The conditions under which this arrangement holds, such as steady operation, startup or an upset. How a departure is detected. Who approves changes.

Six kinds of decision

Ask all six of every decision. Several answers can be yes.

Rule

Does an approved, explicit rule determine the answer, including missing inputs and exceptions?

Estimation

Does this part estimate an unknown quantity, or assign a label, from the evidence available?

Optimization

Is there an agreed objective, known constraints and a model good enough to search the options?

Reasoning

Is the situation outside what the rules and models were built for, so someone must work out what is happening and propose a response?

Coordination

Must two or more people, teams or agents with different objectives agree?

Reserved for people

Does law, policy or the operating arrangement require a person to make or authorize it, or is the evidence too weak to delegate?

Rigor levels

Basic	Limited consequences; errors readily detected and corrected. Lightweight testing; changes recorded.
Enhanced	Material but bounded disruption. Managed versions; material changes tested and approved before use.
High	Approved versions stay fixed until formal management of change approves a change.

Human approval does not by itself make a decision low consequence. Mandatory rules must be implemented and verified in every path an action can take.

DECISION CARD: WORKED EXAMPLE

Deviation triage

"Does this deviation need an investigation, and how deep?" One decision, five kinds of parts, and the path an event takes when an input is missing.

ILLUSTRATIVE EXAMPLE, NOT A VALIDATED PROCEDURE

DECISION ID	ACCOUNTABLE OWNER	RIGOR LEVEL	VERSION	REVIEW DATE
QA-DEV-01	Site head of quality	☐ Basic ☐ Enhanced ■ High	0.1 draft	Set at approval

Rule Site procedure routes the event by type and recorded facts › **Estimation** Validated model estimates likely quality impact and repeat status › **Reasoning** Language model drafts the investigation summary from the batch record › **Reserved** Quality reviewer decides the classification › **Record** Inputs, versions, outputs and the decision kept

Path with a missing input. An event arrives without the environmental data the procedure needs. The rule cannot be evaluated, so it does not count as a pass: the event goes to manual triage with the reason recorded, and no recommendation is made.

1 Decision problem

Allowed answers	Record only; standard investigation; full investigation (as defined in the site procedure)
Trigger, frequency	A deviation is recorded in the quality system; recurring, high volume
Inputs	Deviation record, batch record, equipment and environmental data, earlier deviations of the same type; as recorded at the time of triage
Uncertain	Root cause and quality impact are unknown at triage; records may be incomplete
Objective, limits	Correct classification, timely closure and repeats caught; within the site procedure and applicable investigation requirements

2 Kinds

Kind	What makes this part	Output	Relies on
Rule	Routing by event type and recorded facts	Proposed category, or "cannot evaluate"	Site procedure, version
Estimation	Validated classification model	Likely impact and repeat flag, with confidence	Model ID and version
Reasoning	Language model in an agent	Draft summary for the reviewer	Prompt and model version
Coordination	Production and engineering input when equipment is involved	Facts and context for the reviewer	Named contributors
Reserved	Quality reviewer; Qualified Person for batch disposition	Classification decision	Site procedure

3 Authority

- System recommends a category and drafts the summary; it does not classify.
- Quality reviewer authorizes the classification; the head of quality can override or escalate.
- Permitted system actions: create drafts, attach evidence, route to the review queue.
- The recommendation influences the reviewer, so influence is assessed even though authority stays with the reviewer.

4 Coordination

- Depends on: batch disposition; equipment maintenance decisions.
- Disagreement between contributors goes to the head of quality.

5 Assurance

- **Permission:** every recommendation recorded with its inputs, procedure, model and prompt versions, outputs and the reviewer's decision.
- **Performance:** agreement with reviewers, repeat deviations and closure time, against a baseline and targets agreed before the trial.
- Fallback: manual triage when a model is unavailable or an input is missing.
- High rigor: approved versions fixed; changes through management of change.

6 Operating regime

- Holds for routine production deviations of listed types.
- Events flagged critical at intake, and new products, go straight to a reviewer.
- Changes approved by the head of quality through change control.